Class-II true-increment determinant reduction

TECT verification-first repository · claim A11-CLASSII-TRUE-INCREMENT-DETERMINANT-REDUCTION

first issued 2026-07-21 · this version issued 2026-07-21 · v1.0

1. Purpose and scope

Proved here. The direct past-energy upper form proposed in A10 equation (8.4) is false already for the base Gaussian. The replacement variable

$$I_j = Q_j^{\mathrm{fr}} - q_{B(\phi_{j-1})}(D\phi_{j-1})$$

telescopes the actual A7 action exactly. Its conditional Gaussian log-Laplace transform is an exact regularised determinant plus a positive adapted source-square. The source cannot be discarded or controlled by the Hilbert–Schmidt norm alone.

Open boundary. This note does not prove the source-square bound needed to sum that positive term. It also does not prove the stabilised relative log-Laplace estimate for $\theta I_j + \mathcal{C}_j$, the A7 Nelson bound, or an interacting measure. The two load-bearing successors are `A11-CLASSII-ADAPTED-SOURCE-SQUARE-BOUND` followed by `A11-CLASSII-TRUE-INCREMENT-STABILISED-LOG-LAPLACE`.

The authority is the hash-pinned A1 production Gaussian on $\mathbb{T}_L^3$ with $L = 16$, three complex components in the six-real convention, fixed $\rho_{\mathrm{floor}} = 10^{-12}$, $\eta_{\mathrm{shell}} = 0$, and the A7–A10 sharp rectangular-cube dyadic filtration. The complex Fourier covariance is $2A(k)^{-1}$. This factor two, the distinction between one-axis and total derivative covariance, and the torus volume L^3 are retained throughout. The full-production shell mass is 0.260000000009475; the scalar anchor 0.005 is not used.

2 The past-energy upper form is impossible

Let $N_j = 2^j$, let γ_J be the terminal Gaussian, and write

$$E_J = \sum_{j \le J} q_{B(\phi_{j-1})}(D\phi_{j-1}). \tag{2.1}$$

The candidate A10 upper form was

$$\mathbb{E}_\nu E_J \le \alpha_d \mathcal{H}(\nu \mid \gamma_J) + \epsilon_d \mathbb{E}_\nu \|\phi_J\|_6^6 + K_d \mathbb{E}_\nu \|\phi_J\|_4^4 + C_d, \tag{2.2}$$

with constants independent of J.

Set $\nu = \gamma_J$. Then the entropy term vanishes. Marginal consistency identifies ϕ_{j-1} with the cutoff-N_{j-1} Gaussian. The Gaussian integration-by-parts identity already fixed in A7 gives

$$\mathbb{E}_{\gamma_J} q_{B(\phi_{j-1})}(D\phi_{j-1}) = \mathbb{E}_{\gamma_J} t_{\Gamma_{\le j-1}}(B(\phi_{j-1})). \tag{2.3}$$

If e_N denotes this expectation per unit volume, the A6 sharp-cube theorem states

$$\frac{e_N}{N} \longrightarrow \kappa_{\mathrm{II}} = \delta_{\mathrm{cube}} \mathbb{E} W_\varepsilon(X_\infty) > 0. \tag{2.4}$$

Strict positivity follows from positivity of the production Q_{II} and from W_ε being nonzero on an open set of the nondegenerate Gaussian law. Therefore

$$\frac{\mathbb{E}_{\gamma_J} E_J}{L^3 N_J} = \frac{1}{N_J} \sum_{j \le J} e_{N_{j-1}} \longrightarrow \kappa_{\mathrm{II}} > 0, \tag{2.5}$$

because $\sum_{m=0}^{J-1} 2^m = 2^J - 1$.

On the other hand $A(k)^{-1} = O(|k|^{-4})$ in dimension three, so the point covariances C_N converge in trace. For a circular complex Gaussian vector X with eigenvalues c_a of C_N,

$$\mathbb{E}|X|^4 = (\operatorname{Tr} C_N)^2 + \operatorname{Tr} C_N^2, \qquad \mathbb{E}|X|^6 = (\operatorname{Tr} C_N)^3 + 3 \operatorname{Tr} C_N \operatorname{Tr} C_N^2 + 2 \operatorname{Tr} C_N^3. \tag{2.6}$$

Thus the two endpoint moments on the right of (2.2) remain uniformly bounded, while the left side grows like $L^3 \kappa_{\mathrm{II}} N_J$. This is a contradiction for every fixed choice of $\alpha_d, \epsilon_d, K_d, C_d$. We register it as `F-2026-07-21-A10-PAST-ENERGY-UPPER-FORM`.

3 Exact true-increment telescope

Put $x_j = \phi_{j-1}$ and $z_j = \phi_j$. A10 proved

$$Q_j^{\mathrm{fr}} + \mathcal{C}_j = V_j - V_{j-1} + q_{B(x_j)}(Dx_j). \tag{3.1}$$

Define

$$\boxed{I_j := Q_j^{\mathrm{fr}} - q_{B(x_j)}(Dx_j).} \tag{3.2}$$

Subtracting the last term of (3.1) is an exact identity, not an estimate:

$$\boxed{I_j + \mathcal{C}_j = V_j - V_{j-1}.} \tag{3.3}$$

The empty initial scale is fixed by $\phi_0 = 0$, $\Gamma_{\le 0} = 0$, hence $V_0 = 0$. Consequently

$$\boxed{V_J = \sum_{j \le J} (I_j + \mathcal{C}_j).} \tag{3.4}$$

There is no residual E_J in (3.4).

Condition on the past x_j. In the real shell Hilbert space let

$$A_j = M_{B(x_j)}^{1/2} G_j, \quad r_j = M_{B(x_j)}^{1/2} D x_j, \quad T_j = A_j^* A_j, \quad \ell_j = A_j^* r_j. \tag{3.5}$$

For a standard real Gaussian shell coordinate ξ_j,

$$I_j = \frac{1}{2}\{\langle \xi_j, T_j \xi_j \rangle - \operatorname{Tr} T_j\} + \langle \ell_j, \xi_j \rangle. \tag{3.6}$$

4 Exact determinant and the positive source

At finite cutoff $T_j \geq 0$. Diagonalising T_j and completing the Gaussian square gives, for $p > 0$,

$$\begin{aligned}\log \mathbb{E}_j e^{-pI_j} &= \frac{1}{2}\{p \operatorname{Tr} T_j - \log\det(I + pT_j)\} \\ &\quad + \frac{p^2}{2}\langle \ell_j, (I + pT_j)^{-1}\ell_j\rangle \\ &= -\frac{1}{2}\log\det{}_2(I + pT_j) + \frac{p^2}{2}\langle \ell_j, (I + pT_j)^{-1}\ell_j\rangle.\end{aligned} \tag{4.1}$$

The sign of the second line is positive. The scalar inequality $u - \log(1+u) \leq u^2/2$ for $u \geq 0$ yields

$$\log \mathbb{E}_j e^{-pI_j} \leq \frac{p^2}{4}\|T_j\|_{\mathfrak{S}_2}^2 + \frac{p^2}{2}\|\ell_j\|^2. \tag{4.2}$$

The positive source is unavoidable. In one dimension take $T = \tau > 0$ and $\ell = \sqrt{\tau}q$. Its exact contribution is

$$\frac{p^2\tau q^2}{2(1 + p\tau)}. \tag{4.3}$$

For fixed τ, this diverges quadratically as $|q| \to \infty$, while $\|T\|_{\mathfrak{S}_2}$ remains fixed. Hence no bound depending only on the Hilbert–Schmidt cost can replace (4.2). The crude operator inequality that bounds (4.3) by a multiple of the past energy merely recreates the already refuted E_J route.

Here

$$\ell_j = G_j^* B(x_j) D x_j. \tag{4.4}$$

A9 controls $\sum_j \|T_j\|_{\mathfrak{S}_2}^2$ by a terminal quartic. The new load-bearing analytic statement is instead

$$\sum_j \|G_j^* B(P_{\leq j-1}\phi) D P_{\leq j-1}\phi\|_2^2 \leq C_{\mathrm{src}}\|\phi\|_6^6. \tag{4.5}$$

Equation (4.5) is the gate `A11-CLASSII-ADAPTED-SOURCE-SQUARE-BOUND`. The rational coefficient B is not band-limited; a valid proof must use an actual sharp-cube maximal/square-function or multilinear paraproduct theorem, not a polynomial support shortcut. This note does not prove the source-square bound.

5 The relative variable must also change

For $s = 1 - \theta$, define

$$\widetilde{R}_j^\theta := \theta I_j + \mathcal{C}_j. \tag{5.1}$$

Then (3.4) becomes

$$V_J = \sum_j \{sI_j + \widetilde{R}_j^\theta\}. \tag{5.2}$$

The A10 historical variable was $R_j^\theta = \theta Q_j^{\mathrm{fr}} + \mathcal{C}_j$, and

$$\widetilde{R}_j^\theta = R_j^\theta - \theta q_{B(x_j)}(Dx_j). \tag{5.3}$$

Therefore the old relative log-Laplace gate cannot simply be renamed or reused. After (4.5), the next theorem must be stated for $\theta I_j + \mathcal{C}_j$ and is named `A11-CLASSII-TRUE-INCREMENT-STABILISED-LOG-LAPLACE`.

If (4.5) holds and the A9 quartic constant is C_{fr}, the determinant part has the conditional entropy budget

$$\mathbb{E}_\nu s\sum_j I_j \geq -\alpha_f \mathcal{H}(\nu \mid \gamma_J) - \frac{s^2 C_{\mathrm{fr}}}{4\alpha_f}\mathbb{E}_\nu\|\phi_J\|_4^4 - \frac{s^2 C_{\mathrm{src}}}{2\alpha_f}\mathbb{E}_\nu\|\phi_J\|_6^6. \tag{5.4}$$

This is only a conditional consequence of the open source-square gate. The production sextic remainder must be recomputed with the explicit C_{src}; the previously displayed small reserve cannot be presumed to survive.

6 Executable evidence

The primary route recomputes the A6 covariance and current contractions, uses exact complex-Gaussian fourth and sixth moments, checks (3.3) on a genuine three-component spectral field fixture, and verifies (4.1) by two-dimensional deterministic Gauss–Hermite quadrature. It passes 24/24 assertions. Its independently derived values are

$$\kappa_{\mathrm{II}} = 5.42469581748385 \times 10^{-4}, \qquad \frac{\sum_{N<64} e_N}{64} = 5.06948292202567 \times 10^{-4}. \tag{6.1}$$

The non-importing route directly enumerates cube modes, reconstructs all three Pauli current contractions, and uses a separate scalar determinant quadrature. It passes 18/18 assertions and finds

$$\kappa_{\mathrm{II}} = 5.40500145647357 \times 10^{-4}. \tag{6.2}$$

The small finite-reference difference is within the declared cross-route tolerance and has the same strictly positive conclusion.

7 Proof order now fixed

The surviving order is:

1. retain the negative result for the direct E_J upper form;
2. use the exact true-increment determinant (3.4)–(4.2);
3. prove the adapted source-square estimate (4.5), with an explicit constant compatible with the production sextic budget;
4. prove the new relative log-Laplace estimate for (5.1);
5. reassemble the Nelson inequality, and only then revisit construction of the interacting measure.

8 Devil's-advocate review

1. **DISMISSED – perhaps positive E_J helps a lower bound.** The actual endpoint is the shell sum minus E_J. Its positivity has the wrong direction. The base-Gaussian limit (2.5) makes this quantitative.
2. **DISMISSED – the contradiction may confuse density and integral.** Equations (2.4)–(2.5) explicitly restore the volume L^3. Both sides of (2.2) use torus integrals; the common volume does not alter the linear cutoff contradiction.
3. **VALID WITH MITIGATION – the initial telescope can hide a boundary term.** We fix $\phi_0 = 0$ and $\Gamma_{\leq 0} = 0$, hence $V_0 = 0$. For a different initial scale, (3.4) must include V_0.
4. **DISMISSED – the determinant source should be negative.** Completing the square gives the positive sign in (4.1). Two independent Gauss–Hermite routes verify it and a sign flip fails their assertions.

5. **UPHELD – an HS-only estimate is insufficient.** The scalar family (4.3) is an exact counterexample. The adapted source must be retained.

6. **UPHELD – A10's individual L^4 projector bound closes (4.5).** It does not. Equation (4.5) needs an L^6 vector-valued maximal or multilinear estimate with a cutoff-uniform constant.

7. **UPHELD – rational B has finite polynomial Fourier support.** It does not: the fixed positive denominator prevents a singularity but does not make $B(P_{\leq j}\phi)$ band-limited. No support claim is used here.

8. **UPHELD – the old relative gate can be reused unchanged.** Equation (5.3) shows the missing past energy exactly. A new relative theorem is required.

9. **UPHELD as an overclaim – this closes A7 or promotes T5/T6.** It does neither. The honest result is a scoped T4 structural reduction with two explicit open gates. Operator reproduction may later review T5 without enlarging the statement; T6 and T7 are excluded.

9 Result footer

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Claim: A11-CLASSII-TRUE-INCREMENT-DETERMINANT-REDUCTION
Tier: T4 PROVED-STRUCTURAL-REDUCTION@FIXED-FLOOR-FINITE-CUTOFF
Closed: base-Gaussian no-go for the A10 past-energy upper form;
        exact true-increment action telescope; exact noncentral determinant;
        positive-source and Hilbert-Schmidt-only no-go.
Open: A11-CLASSII-ADAPTED-SOURCE-SQUARE-BOUND;
      A11-CLASSII-TRUE-INCREMENT-STABILISED-LOG-LAPLACE.
Not claimed: source-square closure, A7 Nelson closure, an interacting measure,
             floor removal, infinite volume, BCC, T5, T6, or T7.
Primary expected: 24/24 PASS.
Independent expected: 18/18 PASS.
Integrated expected: 58/58 PASS.
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References

1. TECT claim A6-CLASSII-UV-POWER-COUNTING, sharp-cube covariance and positive Class-II ultraviolet slope.

2. TECT claim A9-CLASSII-SMART-PATH-CANCELLATION, arbitrary-source noncentral Gaussian determinant.

3. TECT claim A10-CLASSII-RELATIVE-COMMUTATOR-REDUCTION, exact action mismatch and sharp rectangular-cube filtration.